WORKED SOLUTION

STEP III 1995, Question 12

Probability and Statistics • Radially Symmetric Densities

STEP III 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.22, *A Distribution With No Mean*: https://beyondstep.keithmcnulty.org/ch12/cauchy.html

The key idea

A radially symmetrical density depends on (u, v) only through the distance $r = \sqrt{u^2 + v^2}$ from the origin, so it looks the same after any rotation about the origin. In particular, the probability of landing in a sector with vertex at the origin depends only on the angle of the sector. The event $U/V < k$ is a pair of opposite sectors cut out by the line $u = kv$ and the u-axis, so its probability is just an angle divided by 2π.

Solution

The joint density and radial symmetry

Each of X and Y has density $\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$. Since X and Y are independent, their joint density is the product of the marginal densities:

$$f(x, y) = \phi(x)\,\phi(y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)},$$

for all real x and y, as required.

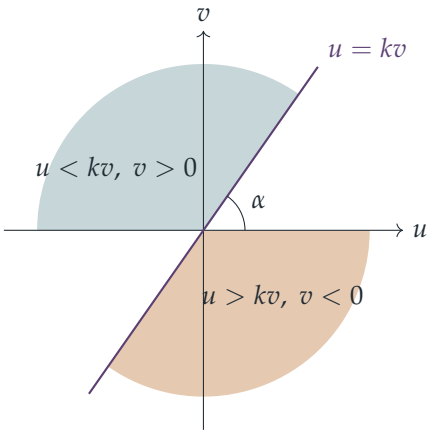
Saying that this density is *radially symmetrical* means that its value at the point (x, y) depends only on the distance $r = \sqrt{x^2 + y^2}$ of the point from the origin, and not on its direction. Here $f(x, y) = \frac{1}{2\pi} e^{-r^2/2}$. Equivalently, the density is unchanged by any rotation of the plane about the origin, and its contours are circles centred on the origin.

The identity for $\mathrm{P}(U/V < k)$

Let U and V have a radially symmetrical joint density $h(u,v) = \rho\big(\sqrt{u^2+v^2}\big)$. The line $v = 0$ has probability zero (it has zero area), so we may ignore the case $V = 0$. Multiplying the inequality $U/V < k$ by V, and reversing it when $V < 0$,

$$\left\{\frac{U}{V} < k\right\} = \{U < kV,\ V > 0\} \cup \{U > kV,\ V < 0\},$$

and the two events on the right are disjoint.



As the diagram shows (drawn for $k > 0$), the line $u = kv$ through the origin and the u-axis split the plane into four sectors, and the two shaded ones make up the event $U/V < k$. The map $(u,v) \mapsto (-u,-v)$ is a rotation through π about the origin, and it takes the region $\{u < kv,\ v > 0\}$ exactly onto $\{u > kv,\ v < 0\}$ (multiply both inequalities by -1). A radially symmetrical density satisfies $h(-u,-v) = h(u,v)$, so these two regions have the same probability. Hence

$$\mathrm{P}\Big(\frac{U}{V} < k\Big) = 2\,\mathrm{P}(U < kV,\ V > 0),$$

which is what we had to show.

The density of U/V

Use polar coordinates $u = r\cos\theta$, $v = r\sin\theta$. The region $\{u < kv,\ v > 0\}$ is the sector $\alpha < \theta < \pi$, where $\alpha \in (0,\pi)$ is the angle the ray $u = kv$, $v > 0$ makes with the positive u-axis, so $\cot\alpha = k$. (Indeed, for $0 < \theta < \pi$ we have $\sin\theta > 0$, and $u < kv$ becomes $\cot\theta < k$; since cot is decreasing on $(0,\pi)$, this is $\theta > \alpha$.)

Ask yourself What is the probability that (U,V) lies in a sector of angle β with vertex at the origin, when we don't know the function ρ?

In polar coordinates the area element is $r\,\mathrm{d}r\,\mathrm{d}\theta$, so a sector $\theta_1 < \theta < \theta_1 + \beta$ has probability

$$\int_{\theta_1}^{\theta_1+\beta}\int_0^\infty \rho(r)\, r\,\mathrm{d}r\,\mathrm{d}\theta = \beta\int_0^\infty \rho(r)\, r\,\mathrm{d}r = \frac{\beta}{2\pi},$$

because taking $\beta = 2\pi$ gives the whole plane, so $2\pi\int_0^\infty \rho(r)\,r\,\mathrm{d}r = 1$. In other words, the direction of (U, V) is uniformly distributed, whatever ρ is.

Our sector has angle $\pi - \alpha$, so

$$\mathrm{G}(k) = \mathrm{P}\Big(\frac{U}{V} < k\Big) = 2\cdot\frac{\pi-\alpha}{2\pi} = 1 - \frac{\alpha}{\pi}.$$

Since $\cot\alpha = k$ with $\alpha \in (0,\pi)$, we have $\alpha = \frac{\pi}{2} - \arctan k$ (both sides lie in $(0,\pi)$ and have cotangent k, because $\cot(\frac{\pi}{2} - \psi) = \tan\psi$). Therefore

$$\mathrm{G}(k) = \frac{1}{2} + \frac{1}{\pi}\arctan k, \qquad -\infty < k < \infty.$$

This rises from 0 to 1, as a distribution function must. Differentiating gives the density of U/V:

$$\mathrm{g}(k) = \mathrm{G}'(k) = \frac{1}{\pi(1+k^2)}, \qquad -\infty < k < \infty,$$

as required. In particular this applies to X/Y for the independent standard normals of the first part.

Checks

- $\mathrm{G}(0) = \frac{1}{2}$: by symmetry U/V is as likely to be positive as negative.
- $\int_{-\infty}^{\infty} \frac{\mathrm{d}k}{\pi(1+k^2)} = \frac{1}{\pi}\big[\arctan k\big]_{-\infty}^{\infty} = 1$.
- A simulation with 2×10^5 points from three different radially symmetrical distributions (two independent standard normals, uniform on the unit disc, and density proportional to e^{-r}) gave $\mathrm{P}(U/V < 0.7) \approx 0.694$ each time, matching $\frac{1}{2} + \frac{1}{\pi}\arctan 0.7 \approx 0.6944$; the two sides of the identity agreed to within sampling error for several values of k.

Common slips

- Writing $\{U/V < k\} = \{U < kV\}$: multiplying by V reverses the inequality when $V < 0$.
- Taking $\alpha = \arctan(1/k)$ without care: this lies in $(-\frac{\pi}{2}, 0)$ when $k < 0$, whereas the angle we need lies in $(0,\pi)$. Writing $\alpha = \frac{\pi}{2} - \arctan k$ works for every k.

Beyond STEP

The density $\frac{1}{\pi(1+k^2)}$ is the standard *Cauchy distribution*. It is famous for having no mean (the integral of $|k|\,\mathrm{g}(k)$ diverges), so the law of large numbers fails for it: the average of n independent Cauchy variables has the same Cauchy distribution as one of them. The polar-coordinate argument here is a first taste of transforming joint densities with a Jacobian, a standard first-year technique. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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